Aquat Ecol
DOI 10.1007/s10452-016-9607-y

Impacts of sediment type on the performance and composition of submerged macrophyte communities

Lin Liu · Xiang-Qi Bu · Jun-Yan Wan · Bi-Cheng Dong · Fang-Li Luo · Hong-Li Li · Fei-Hai Yu

Received: 9 July 2016 / Accepted: 17 October 2016
© Springer Science+Business Media Dordrecht 2016

Abstract To restore deteriorated lake ecosystems, it is important to identify environmental factors that influence submerged macrophyte communities. While sediment is a critical environmental factor for submerged macrophytes and many studies have examined effects of sediment type on the growth of individual submerged macrophytes, very few have tested how sediment type affects the growth and species composition of submerged macrophyte communities. We constructed submerged macrophyte communities containing four co-occurring submerged macrophytes (*Hydrilla verticillata*, *Myriophyllum spicatum*, *Ceratophyllum demersum* and *Chara fragilis*) and subjected them to three sediment treatments, i.e., clay, a mixture of clay and quartz sand at a volume ratio of 1:1 and a mixture at a volume ratio of 1:4. Compared to the clay, the 1:1 mixture treatment greatly increased overall biomass, number of shoot nodes and shoot length of the community, but decreased its diversity. This was because it substantially promoted the growth of *H. verticillata* within the community, making it the most abundant species in the mixture sediment, but decreased that of *M. spicatum* and *C. demersum*. The sediment type had no significant effects on the growth of *C. fragilis*. As a primary nutrient source for plant growth, sediment type can have differential effects on various submerged macrophyte species and 1:1 mixture treatment could enhance the performance of the communities, increasing the overall biomass, number of shoot nodes and shoot length by 39.03%, 150.13% and 9.94%, respectively, compared to the clay treatment. Thus, measures should be taken to mediate the sediment condition to restore submerged macrophyte communities with different dominant species.

Keywords Sediment type · Submerged macrophytes · Community diversity · Species composition

Handling Editor: Kevin Murphy.

L. Liu · X.-Q. Bu · J.-Y. Wan · B.-C. Dong · F.-L. Luo · H.-L. Li (✉) · F.-H. Yu
School of Nature Conservation, Beijing Forestry University, Beijing 100083, China
e-mail: lihongli327@163.com

Introduction

The majority of lakes in China are influenced by eutrophication, and subsequent degradation and deterioration of water quality due to anthropogenic activities (Cheng and Li 2006; Qin et al. 2013). As a consequence, the abundance and diversity of submerged aquatic plant-life communities are negatively impacted (Alahuhta et al. 2013; Bornette and Puijalon 2011). In addition, a gradual shift from the clear-water state, a state dominated by macrophytes, to the turbid-water state, a state featuring by high phytoplankton biomass has often occurred, which breaks the

Published online: 25 October 2016

equilibria of the lake ecosystem (Hilt and Gross 2008; Körner 2002; Qin et al. 2013; Sand-Jensen et al. 2000).

To restore the health of lakes and promote its benefits to humans, the recovery of the aquatic ecosystem, especially the submerged aquatic plant life, is of the upmost importance (Wang et al. 2016; Zhang et al. 2012). Submerged aquatic plant communities are one of the most important components of aquatic ecosystems and serve as key players in multiple ecological functions, such as promoting water purification by increasing water transparency and the retention of potentially polluting nutrients (Bakker et al. 2010; Engelhardt and Ritchie 2001; Hilt and Gross 2008), providing food resources for other aquatic species (Deegan and Ganf 2008; Reid et al. 2008; Watson and Barmuta 2011), and forming a reasonable ecological structure within the community (Antunes et al. 2012; Mitsch et al. 2005). Hence, the sustainable management and efficient restoration of submerged macrophytes are critically important (Peretyatko et al. 2009).

Submerged macrophyte communities can be affected by many factors, biotic or abiotic in nature, such as sediment characteristics, nutrient limitation, light availability, seasonal changes, hydromorphological changes, plant growth type and relationships with fish and waterfowl (Alahuhta et al. 2013; Bornette and Puijalon 2011; Irfanullah and Moss 2004; Li et al. 2015; Rodrigo et al. 2013; Wang et al. 2016). Among these factors, nutrient availability is considered as one of the most important environmental factors which influences the growth of submerged aquatic macrophytes (Jiang et al. 2008; Xie and Yu 2011). Sediment was assumed to be the primary nutrient source (e.g., nitrogen, phosphorus or manganese) for plant growth in most cases (Hussner et al. 2009) due to the fact that the absorbable nutrient content in sediment is much higher than that in the water column (Carr 1998).

Substantial studies have investigated the effects of sediment type on the growth and succession of specific submerged macrophyte species, such as *Vallisneria natans* (Bai et al. 2014; Li et al. 2012; Xiao et al. 2007), *Hydrilla verticillata* (Wu et al. 2009; Zhu et al. 2012), *Myriophyllum spicatum* (Jiang et al. 2008; Wang et al. 2009) and *Potamogeton crispus* (Jiang et al. 2008; Wang 2013), and most of them explored the responses of root morphology to sediment type. These effects could be related to the organic matter content, nutrient level and texture of sediment (Bai et al. 2014; Bornette and Puijalon 2011; Li et al. 2012). Furthermore, sediment that contains a greater proportion of clay has more organic matter and greater pore water nutrient concentrations and shows reductions in oxygen exchange with the water column, which result in anoxic sediment conditions (Jarvis and Moore 2015). The effects of sediment may impact submerged plant species differently, which in turn affects the composition of the community (Johnson and Ostrofsky 2004). However, few studies have tested the community-level effects of sediment type on species composition and biomass, especially of submerged macrophyte communities. An enhanced understanding of these response mechanisms to sediment type will provide useful information for the management of aquatic vegetation.

To examine the effects of sediment type on the performance and species composition of submerged macrophyte communities, we assembled experimental aquatic communities consisting of four co-occurring submerged macrophytes and grew them in three sediment types of clay, a mixture of clay and quartz sand at a volume ratio of 1:1 and a mixture at a ratio of 1:4. Specifically, we addressed the following questions. (1) Does the sediment type affect the performance (measured as biomass, shoot node number and shoot length) and species diversity of submerged macrophyte communities? (2) Does the sediment type affect the performance (measured as biomass, node number and shoot length) of each individual species? (3) Can the changes at the community level be explained by the responses of the individual species?

Methods

Materials

Experimental communities assembled in this study were composed of four co-occurring, widely distributed, submerged macrophytes found in China, i.e., *Hydrilla verticillata* R. (Hydrocharitaceae), *Myriophyllum spicatum* L. (Haloragidaceae), *Ceratophyllum demersum* L. (Ceratophyllaceae) and *Chara fragilis* Desv. (Characeae). All selected plants are perennial species capable of clonal growth; specifically, their stem can develop into an integral individual (Sosnová et al. 2010, 2011; Zhang et al. 2014), and they can coexist in natural habitats. All these species

are common and can be easily collected in lakes in Beijing, which contributed to their selection for this study.

In mid-June of 2012, stem fragments of *M. spicatum*, *C. demersum* and *C. fragilis* were collected from the lakes of the Winter Palace, and those of *H. verticillata* were from the lakes of the Beijing Olympic Forestry Park in Beijing. Plants of each species were collected from at least two lakes. All four species were vegetatively propagated in a greenhouse at the Forestry Science Co, Ltd. of Beijing Forestry University for one month to minimize the potential effect of local environments.

On July 14, 2012, 92 shoots of each species were selected, and each was trimmed to a length of 15 cm, with an apical bud but no lateral branches, to minimize variation in the initial size. Of the 92 shoot fragments chosen for each species, 20 were randomly selected for initial biomass measurements, and the other 72 were used for the experiment.

Experimental design

Experiment communities were set up in 18 plastic boxes (length 34 cm; width 34 cm; and depth 53 cm), each containing an 18-cm-deep layer of one of the three sediment treatments. The three sediment types were clay, a mixture of clay and quartz sand at a volume ratio of 1:1 and a mixture of clay and quart sand at a volume ratio of 1:4 (Fig. 1). Each treatments had six replicates. The clay (1.36 ± 0.14 mg total N g^{-1} soil [mean $\pm$ SE], 0.18 ± 0.02 mg total P g^{-1}, 11.43 ± 0.34 mg total K g^{-1}) was collected from the Cui National Urban Wetland Park in Beijing. Before planting, all boxes were filled with tap water to a depth of 20 cm above the soil surface. Four shoot fragments of each of the four species were randomly planted in 16 fixed positions according to randomization generator, which were arranged in four rows and four columns in each box, following a Latin square approach, i.e., the four shoot fragments of each species were distributed in the four rows, with one shoot in each rows and never 2 fragments of the same species in the same column (Fig. 1). In this study, all shoot fragments were planted and anchored into the sediment, despite the rootless nature of *C. demersum* in natural conditions (Xue et al. 2012). Immediately after planting, the boxes were filled with tap water to 55 cm above the sediment surface.

The duration of the experiment was 2 months, ending in mid-September. Biomass and species composition of submerged macrophyte communities can change even within a short time period (Wang et al. 2008; Zhang et al. 2012), making the experiment viable. During the experiment, tap water (0.79 ± 0.06 mg L^{-1} TN, 0.012 ± 0.0005 mg L^{-1} TP) was added to the boxes every day to maintain the water level at 55 cm above the sediment. The nutrient concentrations of tap water did not change during the experiment. During the experiment, no plant fragments were removed from the box, thus the final biomass was the total biomass for the whole box. The temperature in the greenhouse was 25.5 ± 0.3 °C and was measured hourly using two Hygrochron temperature/humidity loggers (iButton DS1923; Maxim Integrated Products, USA). The light intensity in the greenhouse was 467 ± 53 μmol m^{-2} s^{-1} and was measured on sunny days during the experiment. During the experiment, any algae that were clearly visible in the boxes were manually cleared out using a small, circular fishnet (diameter 9 cm; mesh size 0.5 mm).

Harvest and measurements

In mid-September of 2012, all the surviving aquatic macrophytes plants were carefully harvested with their roots and then sorted into species. Number of shoot nodes, which is a measure of potential clonal growth because every single node can potentially develop into a new plant, and total shoot length, which is a measurement of local vegetative spread (Zhang et al. 2014), were determined during the harvest. A subsample of five shoot fragments for each species was collected from each experimental community to count number of nodes and measure shoot length. Biomass of these samples was measured separately. The remaining parts of each species in each community were oven-dried at 70 °C for 72 h and then weighed.

Data analyses

Based on the data from the subsample of the five fragments per species, number of shoot nodes per unit biomass and shoot length per unit biomass was calculated for each species in each community (Zhang et al. 2014). Then, total number of shoot nodes for each species in each community (i.e., in each box) was

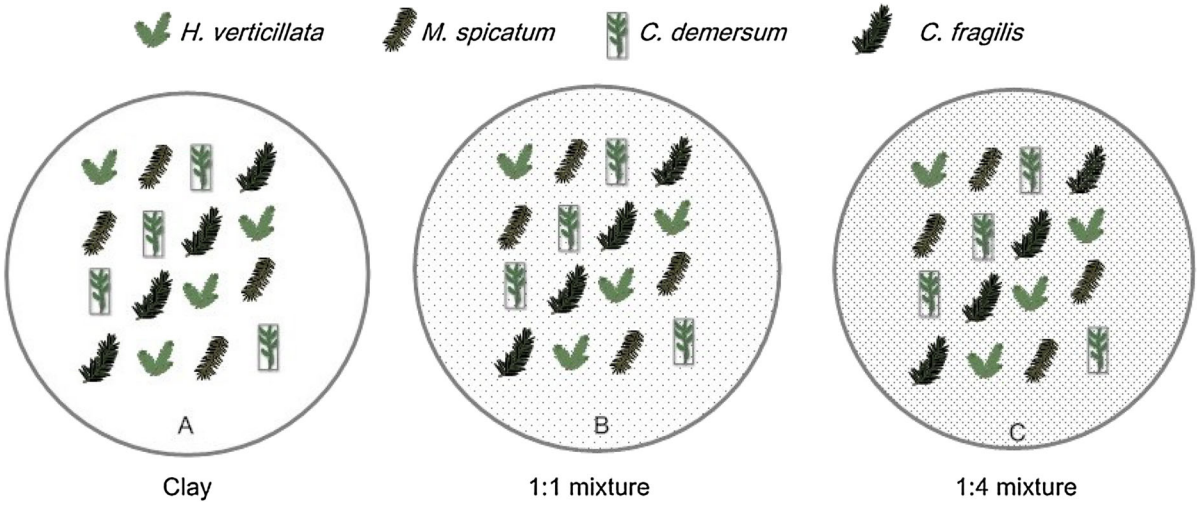

Fig. 1 Experimental design. There were three sediment treatments: **a** clay which indicated the sediment was clay; **b** 1:1 mixture indicated that the sediment was a mixture of clay and quartz sand at a volume ratio of 1:1; **c** 1:4 mixture indicated the sediment was a mixture of clay and quartz sand at a volume ratio of 1:4. Each treatment had six replicates

derived by multiplying biomass with number of nodes per unit biomass. Similarly, total shoot length for each species in each box was calculated. Shannon–Wiener diversity index (H) was calculated as: $H = -\sum P_i \ln(P_i)$ ($i = 1, 2 \ldots S$), where S is number of macrophyte species and P_i is the growth parameter (biomass, number of nodes or shoot length) of the species i divided by the sum of the growth measurement for all four species in the community (Kent and Coker 1992). Because the species number did not change, an increase in diversity indices suggests that the four species are distributed more evenly in the community, and a decrease suggests that they are distributed less evenly. Furthermore, the proportions of each species in the community were calculated as the ratio of biomass, number of nodes and shoot length of each species to that of the community.

All data analyses were conducted using SPSS 19.0 (IBM Corp., Armonk, NY, USA). We performed one-way ANOVAs to examine the effects of sediment type on the overall growth (total biomass, number of nodes and shoot length) and Shannon–Wiener diversity index of the macrophyte communities and on the three growth parameters and proportion for each of the four species individually. Post hoc comparisons were done using Tukey HSD tests. Before the analyses, proportions of biomass, node number and shoot length of both *H. verticillata* and *C. fragilis* and proportions of biomass and shoot length of *M. spicatum* were transformed to arcsin to reduce heteroscedasticity.

Results

Effects of sediment type on the community

Biomass, number of nodes and shoot length of the submerged macrophyte communities were or tended to ($P < 0.1$) the greatest (7.3017 ± 0.6806 g, 2.4920 × 10^3 ± 0.2768 × 10^3 and 33.7967 ± 3.3123 m, respectively) in the 1:1 mixture of clay and quartz sand, smallest (5.2520 ± 0.4439 g, 0.9963 × 10^3 ± 0.1577 × 10^3 and 17.7000 ± 2.0457 m, respectively) in the clay sediment and intermediate in the 1:4 mixture of clay and quartz sand (Fig. 2a–c). The diversity indices of number of nodes and shoot length showed exactly the opposite pattern: they were the smallest in the 1:1 mixture sediment, greatest in the clay sediment and in-between in the 1:4 mixture sediment (Fig. 2e, f). The diversity index of biomass did not differ significantly among the three sediment treatments (Fig. 2d).

Effects of sediment type on growth of individual species

The four submerged macrophyte species in the communities responded differently to sediment type (Fig. 3). For *H. verticillata*, the measured responses were similar to those of the total community: all three growth measures were the greatest in the 1:1 mixture sediment, smallest in the clay sediment and intermediate in the 1:4 mixture sediment (Fig. 3a–c). For *M.*

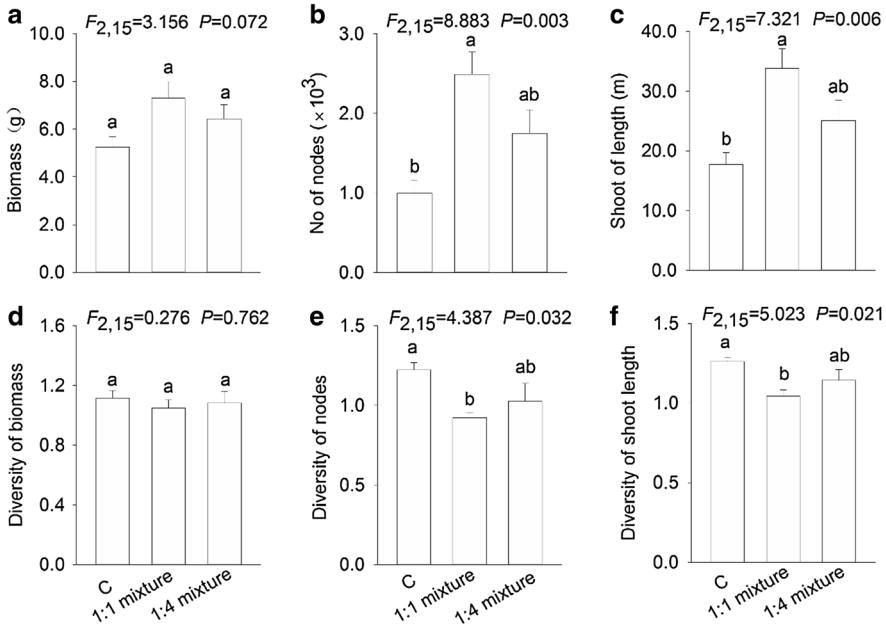

Fig. 2 Effects of sediment type on the growth (**a**–**c**) and diversity (**d**–**f**) of the submerged communities. *Bars* and *vertical lines* are means + 1 SE. *Bars sharing the same letters* are not significantly different at $P = 0.05$. *F* and *P* values of the ANOVA test are given. Treatment codes: C = clay as the sediment, 1:1 mixture = sediment with a mixture of clay and quartz sand at a volume ratio of 1:1, and 1:4 mixture = sediment with a mixture of clay and quartz sand at a volume ratio of 1:4

spicatum and *C. demersum*, biomass in the clay sediment was significantly greater than that in the 1:1 and 1:4 mixture treatments (Fig. 3d, g), but number of nodes and shoot length did not differ significantly among the three sediment types (Fig. 3e, f, h, i). Sediment type had little effect on the growth of *C. fragilis* (Fig. 3j–l).

Effects of sediment type on species proportion

Sediment type significantly affected the proportion of biomass, node number and shoot length for each of the four species in the communities (Fig. 4). For *H. verticillata*, all three proportions were significantly higher in the two mixture treatments than in the clay sediment. The proportions of biomass, nodes number and shoot length of *H. verticillata* were about 50% (49.21, 54.91 and 48.66%, respectively) in the 1:4 mixture sediment and were much over 50% (58.14, 67.82 and 62.89%) in the 1:1 mixture sediment, suggesting that *H. verticillata* dominated the submerged macrophyte communities in the two mixture condition (Fig. 4). The proportions of biomass of *M. spicatum* and node number and shoot length of *C. demersum* were all significantly greater in the clay than in the mixture treatments (Fig. 4). However, the proportions of node number and shoot length of *M. spicatum* and biomass of *C. demersum* were significantly greater in the clay than in the 1:1 mixture treatment and were not significantly different between the clay and the 1:4 mixture treatment (Fig. 4). The proportion of biomass for *C. fragilis* did not differ among these three sediments. The proportions of node number and shoot length for this species were significantly greater in the clay than in the 1:1 mixture treatment, but they did not differ between the clay and the 1:4 mixture treatment (Fig. 4).

Discussion

Results suggest that the 1:1 mixture of clay and quartz sand could dramatically enhance the growth and reduce the diversity of the submerged macrophyte communities compared to the clay treatment. Several studies have shown that, as a primary nutrient source, sediment could exert a significant effect on the growth of rooted submerged macrophytes (Carling et al. 2013; Hussner et al. 2009; Xie et al. 2005). As quartz sand has no nutrients, when its proportion increased, the nutrient-rich clay content decreased in the sediment and thus there was a decrease in nutrient availability for plant growth. However, we found an increase in the growth of submerged macrophyte communities when they were grown in a 1:1 mixture of clay and quartz sand, suggesting that the higher nutrient content of

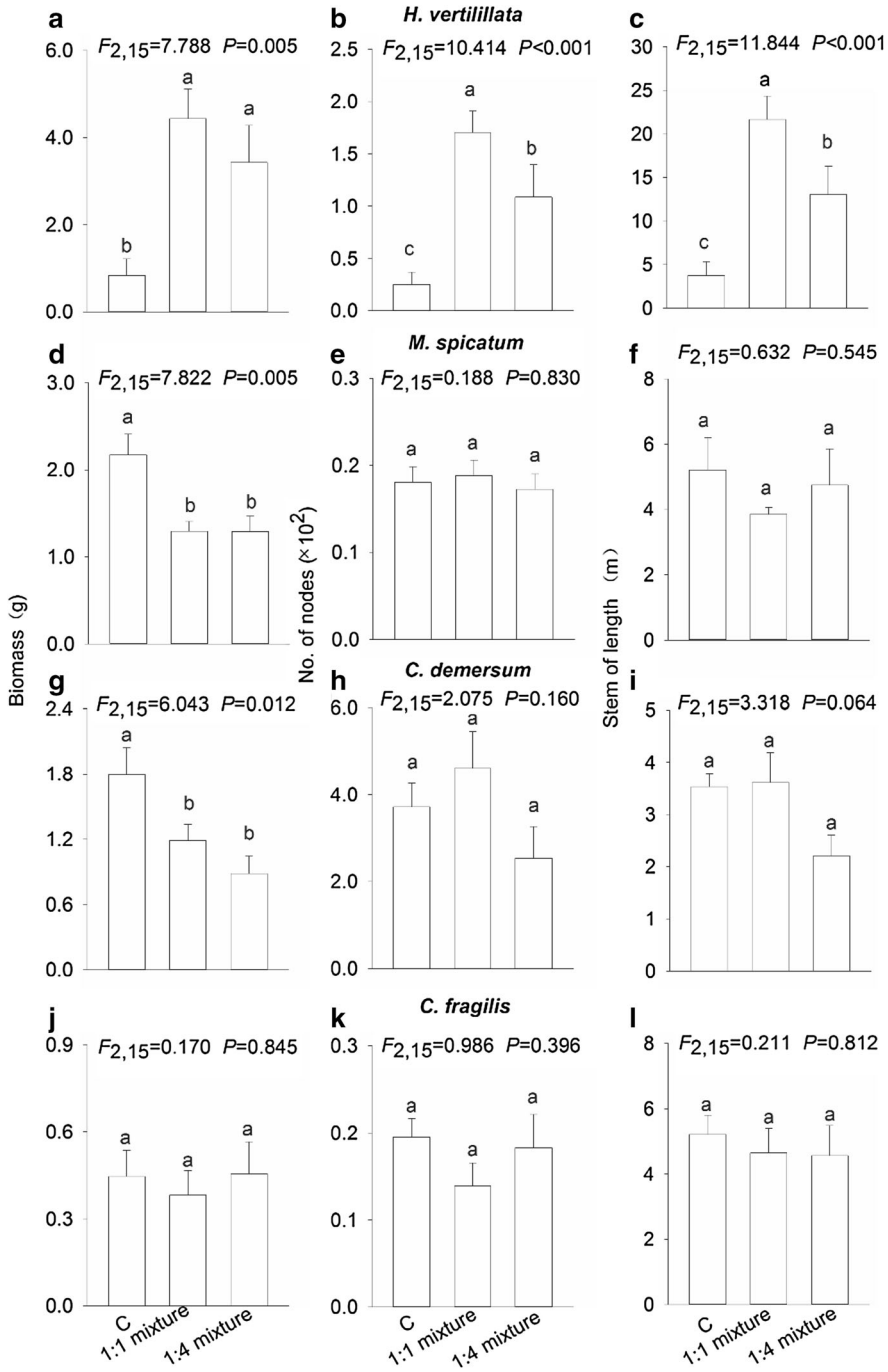

Fig. 3 Effects of sediment type on growth measures of each of the four submerged species. *Bars* and *vertical lines* are means ± 1 SE. *Bars sharing the same letters* are not significantly different at $P = 0.05$. F and P values of the ANOVA tests are given. Treatment codes are described as in Fig. 2

sediment could inhibit the growth of macrophyte communities. Also, colloidal compounds of clay could inhibit the growth via physical effects on water flow through root (Asli and Neumann 2009). Moreover, the sediment would change the performance (such as, biomass, node number and shoot length) of abundant species, which usually affect the performance of plant communities (Zhang et al. 2014). Zhang et al. (2014) found that the reduction in richness of the dominant species could contribute to the decrease in the performance of submerged macrophyte communities. In the study, adding quartz sand to the substrate

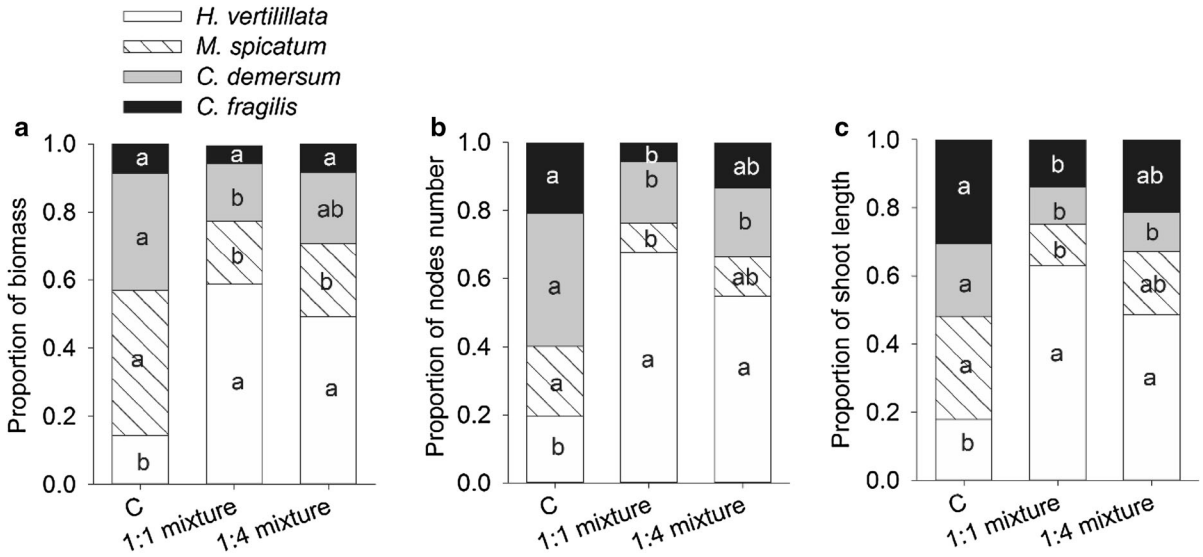

Fig. 4 Effects of sediment type on species composition of the submerged communities. *Bars* are means and those sharing the *same letters* are not significantly different at $P = 0.05$. Treatment codes are described as in Fig. 2

resulted in the change of the dominant species from *M. spicatum* to *H. verticillata*. Therefore, the increased biomass of the community was mainly due to the markedly increased biomass of *H. verticillata*.

The growth and proportion of the four submerged macrophyte species responded differently to changes in the sediment. In the clay sediment, which had the highest nutrient contents, *M. spicatum* was the dominant species and produced the largest biomass among the four species, consistent with previous studies (Xie et al. 2013). Previous studies have also illustrated that *M. spicatum* prefers to grow in sediments that have higher organic content (Kimbel 1982; Xiao et al. 2006). As a competitive strategy species (Murphy et al. 1990), *M. spicatum* grows rapidly after establishing in a favorable environment and reaches a large size to capture more light, which leads to a reduction in soil nutrient uptake by neighboring plants (Smith and Barko 1990). It also tends to develop greater branches and form dense beds (Xie et al. 2013). The dense growth of *M. spicatum* partly led to the clear suppression of *H. verticillata*.

This suppression may also be a result of the sensitivity of *H. verticillata* to low light (Mudge and Haller 2012) for its facultative C_4 NADP-malic enzyme system (Rao et al. 2006). *H. verticillata* has aerenchyma tissues that allow oxygen to be transported to below-ground tissues to support root respiratory processes (Wu et al. 2009). The organic enrichments of sediment are usually followed by various chemical and biological processes, which consume a large amount of sediment oxygen and result in release of phytotoxins (such as methane and sulfur dioxide) (Silveira and Thomaz 2015; Wijck et al. 1992). Because *H. verticillata* was sensitive to phytotoxins in below-ground water, the performance of *H. verticillata* declines in clay (Wu et al. 2009, 2015). However, in the mixed-soil treatments, *H. verticillata* produced 3–12 times more biomass, nodes and shoot length than the other three less abundant species (*C. demersum*, *M. spicatum* and *C. fragilis*). The proportions of biomass, node number and shoot length of *H. verticillata* were about 50% of the submerged macrophyte communities in the mixed-soil treatments. Therefore, *H. verticillata* was the superior competitor in the submerged macrophyte communities in the latter two cases, where the sediment contains low to medium concentrations of organic matter and nutrients such as nitrogen and phosphorus. This result is in agreement with the findings of Silveira and Thomaz (2015). The rapid growth and dominance of *H. verticillata* belie its low light compensate point or low light requirement for photosynthesis (Haller and Sutton 1975; Van et al. 1976) and its high adaptation to low nutrient level (Cook and Lüönd 1982). Anchored in a limited underwater light environment, submerged

species such as *H. verticillata* with a low light compensate point would succeed in competing for limited light, which enable those species to rapidly absorb nutrients and grow better (Yan et al. 1997).

Also, *H. verticillata* can develop a dense light excluding upper-canopy at the water surface, almost completely eliminating sunlight penetration, and thereby limiting competition from other submerged species that grow at lower depths (Haller and Sutton 1975; Hussner et al. 2014). This aquatic plant can also develop a greater deal of lateral branches in this condition, whose fragments contribute to plant dispersion and further enhance the dominance of this species (Silveira and Thomaz 2015). Therefore, *H. verticillata* would inhibit the growth of other species in submerged macrophyte communities from light and nutrient. Because it occurred in the low light conditions under the shade of *H. verticillata*, the decreased biomass of *M. spicatum* may be explained by its decrease in uptake phosphate ability in low light condition (Zhang et al. 2011) and growth of *C. demersum* was inhibited probably by shortage of carbohydrate production (Cao et al. 2011). *C. fragilis* is less sensitive to light, and therefore, it could maintain its growth despite the shading of the abundant species (Li et al. 2015).

Changes in species diversity, composition and biomass of the communities in response to the various sediments could partly be a result of the differential responses of individual species (Smith and Knapp 2003). Also, sediment may shift the competitive balance among the co-occurring submerged aquatic plant species and thus modify the growth and structure of the communities (Heil and Bruggink 1987; Yu et al. 2010). For example, previous investigations have found that sediment can modify interactions between *H. verticillata* and *Egeria densa* (Mony et al. 2007) and between *Elodea nuttallii* and *M. spicatum* (Angelstein et al. 2009). Maybe sediment type can greatly alter the performance and species composition of submerged macrophyte communities and that the fertile sediment can hinder the growth of the submerged community. Hence, the sediment condition such as composition and nutrients is vital for the restoration of submerged macrophytes communities with different dominant species in a degraded shallow lake.

Although the effect of sediment on the growth of macrophytes may be primarily via nutrient level, geochemical conditions such as redox potential may also have some effects (Ye et al. 2009). The experiment condition in boxes in this study simulates field condition at a shallow water level, but results may differ from natural conditions in deeper water where a more reductive condition in sediment is expected (Ye et al. 2009). Further studies must be completed to elucidate the influence of sediment on macrophyte growth and community structure at a deeper water level. Moreover, the effects of sediment changes on species interactions could not be verified by the present experiment. To separate the effects of sediments on changes in the growth of individual species or changes in the competitive interactions among species, it is essential to further construct monocultures containing a single species.

Acknowledgements This study was funded by the Fundamental Research Funds for the Central Universities (2015ZCQ-BH-01), the National Natural Science Foundation of China (31470475) and the National key Research and Development Program (2016YFC1201100).

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

References

Alahuhta J, Kanninen A, Hellsten S, Vuori KM, Kuoppala M, Hämäläinen H (2013) Environmental and spatial correlates of community composition, richness and status of boreal lake macrophytes. Ecol Indic 32:172–181

Angelstein S, Wolfram C, Rahn K, Kiwel U, Frimel S, Merbach I, Schubert H (2009) The influence of different sediment nutrient contents on growth and competition of *Elodea nuttallii* and *Myriophyllum spicatum* in nutrient-poor waters. Fund Appl Limnol 175:49–57

Antunes C, Correia O, Silva JMD, Cruces A, Freitas MDC, Branquinho C (2012) Factors involved in spatiotemporal dynamics of submerged macrophytes in a Portuguese coastal lagoon under Mediterranean climate. Estuar Coastal Shelf Sci 110:93–100

Asli S, Neumann PM (2009) Colloidal suspensions of clay or titanium dioxide nanoparticles can inhibit leaf growth and transpiration via physical effects on root water transport. Plant Cell Environ 32:577–584

Bai X, Chen K, Zhao H, Chen X (2014) Impact of water depth and sediment type on root morphology of the submerged plant *Vallisneria natans*. J Freshw Ecol 30:75–84

Bakker ES, Donk EV, Declerck SAJ, Helmsing NR, Hidding B, Nolet BA (2010) Effect of macrophyte community composition and nutrient enrichment on plant biomass and algal blooms. Basic Appl Ecol 11:432–439

Bornette G, Puijalon S (2011) Response of aquatic plants to abiotic factors: a review. Aquat Sci 73:1–14

Cao TE, Leyi NI, Xie P, Jun XU, Zhang M (2011) Effects of moderate ammonium enrichment on three submersed macrophytes under contrasting light availability. Freshw Biol 56:1620–1629

Carling GT et al (2013) Relationships of surface water, pore water, and sediment chemistry in wetlands adjacent to Great Salt Lake, Utah, and potential impacts on plant community health. Sci Total Environ 443:798–811

Carr GM (1998) Macrophyte growth and sediment phosphorus and nitrogen in a Canadian prairie river. Freshw Biol 39:525–536

Cheng X, Li S (2006) An analysis on the evolvement processes of lake eutrophication and their characteristics of the typical lakes in the middle and lower reaches of Yangtze River. Chin Sci Bull 51:1603–1613

Cook CDK, Lüönd R (1982) A revision of the genus *Hydrocharis* (hydrocharitaceae). Aquat Bot 14:177–204

Deegan BM, Ganf GG (2008) The loss of aquatic and riparian plant communities: implications for their consumers in a riverine food web. Austral Ecol 33:672–683

Engelhardt KA, Ritchie ME (2001) Effects of macrophyte species richness on wetland ecosystem functioning and services. Nature 411:687–689

Haller WT, Sutton DL (1975) Community structure and competition between *Hydrilla* and *Vallisneria*. Hyacinth Control J 13:48–50

Heil GW, Bruggink M (1987) Competition for nutrients between *Calluna vulgaris* (L.) Hull and *Molinia caerulea* (L.) Moench. Oecologia 73:105–107

Hilt S, Gross EM (2008) Can allelopathically active submerged macrophytes stabilise clear-water states in shallow lakes? Basic Appl Ecol 9:422–432

Hussner A, Meyer C, Busch J (2009) The influence of water level and nutrient availability on growth and root system development of *Myriophyllum aquaticum*. Weed Res 49:73–80

Hussner A, Hofstra D, Jahns P, Clayton J (2014) Response capacity to CO_2 depletion rather than temperature and light effects explain the growth success of three alien Hydrocharitaceae compared with native *Myriophyllum triphyllum* in New Zealand. Aquat Bot 120:205–211

Irfanullah HM, Moss B (2004) Factors influencing the return of submerged plants to a clear-water, shallow temperate lake. Aquat Bot 80:177–191

Jarvis JC, Moore KA (2015) Effects of seed source, sediment type, and burial depth on mixed-annual and perennial *Zostera marina* L. seed germination and seedling establishment. Estuar Coasts 38:964–978

Jiang J, Zhou C, An S, Yang H, Guan B, Cai Y (2008) Sediment type, population density and their combined effect greatly charge the short-time growth of two common submerged macrophytes. Ecol Eng 34:79–90

Johnson RK, Ostrofsky ML (2004) Effects of sediment nutrients and depth on small-scale spatial heterogeneity of submersed macrophyte communities in Lake Pleasant, Pennsylvania. Can J Fish Aquat Sci 61:1493–1502

Kent M, Coker P (1992) Vegetation description and analysis. Area 12:76–77

Kimbel JC (1982) Factors influencing potential intralake colonization by *Myriophyllum spicatum* L. Aquat Bot 14:295–307

Körner S (2002) Loss of submerged macrophytes in shallow lakes in north-eastern Germany. Int Rev Gesamten Hydrobiol Hydrogr 87:375–384

Li ZQ et al (2012) Effect of substrate grain size on the growth and morphology of the submersed macrophyte *Vallisneria natans* L. Limnol Ecol Manag Inland Waters 42:81–85

Li HL, Xu YS, Wang YY, Yu NQ, Zhang MX, Lei GC, Yu FH (2015) Does clonal fragmentation of the floating plant *Eichhornia crassipes* affect the growth of submerged macrophyte communities? Folia Geobot 50:283–291

Mitsch WJ, Li Z, Anderson CJ, Altor AE, Hernández ME (2005) Creating riverine wetlands: ecological succession, nutrient retention, and pulsing effects. Ecol Eng 25:510–527

Mony C, Koschnick TJ, Haller WT, Muller S (2007) Competition between two invasive Hydrocharitaceae (*Hydrilla verticillata* (L.f.) (Royle) and *Egeria densa* (Planch)) as influenced by sediment fertility and season. Aquat Bot 86:236–242

Mudge CR, Haller WT (2012) The influence of pH and light on hydrilla (*Hydrilla verticillata*) photosynthesis and chlorophyll after exposure to flumioxazin. Weed Sci 60:4–9

Murphy KJ, Rørslett B, Springuel I (1990) Strategy analysis of submerged lake macrophyte communities: an international example. Aquat Bot 36:303–323

Peretyatko A, Teissier S, Backer SD, Triest L (2009) Restoration potential of biomanipulation for eutrophic peri-urban ponds: the role of zooplankton size and submerged macrophyte cover. Hydrobiologia 634:125–135

Qin B et al (2013) Lake eutrophication and its ecosystem response. Chin Sci Bull 58:961–970

Rao S, Reiskind J, Bowes G (2006) Light regulation of the photosynthetic phosphoenolpyruvate carboxylase (PEPC) in *Hydrilla verticillata*. Plant Cell Physiol 47:1206–1216

Reid DJ, Quinn GP, Lake PS, Reich P (2008) Terrestrial detritus supports the food webs in lowland intermittent streams of south-eastern Australia: a stable isotope study. Freshw Biol 53:2036–2050

Rodrigo MA, Rojo C, Alonso-Guillén JL, Vera P (2013) Restoration of two small Mediterranean lagoons: the dynamics of submerged macrophytes and factors that affect the success of revegetation. Ecol Eng 54:1–15

Sand-Jensen K, Tenna R, Vestergaard O, Larsen SE (2000) Macrophyte decline in Danish lakes and streams over the past 100 years. J Ecol 88:1030–1040

Silveira MJ, Thomaz SM (2015) Growth of a native versus an invasive submerged aquatic macrophyte differs in relation to mud and organic matter concentrations in sediment. Aquat Bot 124:85–91

Smith CS, Barko JW (1990) Ecology of Eurasian watermilfoil. J Aquat Plant Manage 28:55–64

Smith MD, Knapp AK (2003) Dominant species maintain ecosystem function with non-random species loss. Ecol Lett 6:509–517

Sosnová M, Diggelen RV, Klimešová J (2010) Distribution of clonal growth forms in wetlands. Aquat Bot 92:33–39

Sosnová M, Diggelen RV, Macek P, Klimešová J (2011) Distribution of clonal growth traits among wetland habitats. Aquat Bot 95:88–93

Van TK, Haller WT, Bowes G (1976) Comparison of the photosynthetic characteristics of three submersed aquatic plants. Plant Physiol 58:761–768

Wang L (2013) Response of *Potamogeton crispus* root characteristics to sediment heterogeneity. Acta Ecol Sin 33:282–286

Wang J, Yu D, Wang Q (2008) Growth, biomass allocation, and autofragmentation responses to root and shoot competition in *Myriophyllum spicatum* as a function of sediment nutrient supply. Aquat Bot 89:357–364

Wang S, Jin X, Jiao L, Wu F (2009) Response in root morphology and nutrient contents of *Myriophyllum spicatum* to sediment type. Ecol Eng 35:1264–1270

Wang P, Zhang Q, Xu YS, Yu FH (2016) Effects of water level fluctuation on the growth of submerged macrophyte communities. Flora Morphol Distrib Funct Ecol Plants 223: 83–89

Watson A, Barmuta LA (2011) Feeding-preference trials confirm unexpected stable isotope analysis results: freshwater macroinvertebrates do consume macrophytes. Mar Freshw Res 62:1248–1257

Wijck CV, Groot CJD, Grillas P (1992) The effect of anaerobic sediment on the growth of *Potamogeton pectinatus*: the role of organic matter, sulphide, and ferrous iron. Aquat Bot 44:31–49

Wu J, Cheng S, Liang W, Wu Z (2009) Effects of organic-rich sediment and below-ground sulfide exposure on submerged macrophyte, *Hydrilla verticillata*. Bull Environ Contam Toxicol 83:497–501

Wu J, Dai YR, Rui SY, Cui NX, Zhong F, Cheng SP (2015) Acclimation of Hydrilla verticillata to sediment anoxia in vegetation restoration in eutrophic waters. Ecotoxicology 24:2181–2189

Xiao DR, Tian K, Yuan H, Yang YM, Li NY, Xu SG (2006) The distribution patterns and changes of aquatic plant communities in Napahai Wetland in northwestern Yunnan Plateau, China. Acta Ecol Sin 3:338–343

Xiao K, Yu D, Wu Z (2007) Differential effects of water depth and sediment type on clonal growth of the submersed macrophyte *Vallisneria natans*. Hydrobiologia 589:265–272

Xie D, Yu D (2011) Size-related auto-fragment production and carbohydrate storage in auto-fragment of *Myriophyllum spicatum* L. in response to sediment nutrient and plant density. Hydrobiologia 658:221–231

Xie Y, An S, Yao X, Xiao K, Zhang C (2005) Short-time response in root morphology of *Vallisneria natans* to sediment type and water-column nutrient. Aquat Bot 81:85–96

Xie D, Yu D, You WH, Wang LG (2013) Morphological and physiological responses to sediment nutrients in the submerged macrophyte *Myriophyllum spicatum*. Wetlands 33:1095–1102

Xue P, Yan C, Sun G, Luo Z (2012) Arsenic accumulation and speciation in the submerged macrophyte *Ceratophyllum demersum* L. Environ Sci Pollut R Int 19:3969–3976

Yan G, Ma J, Qiu D, Wu Z (1997) Succession and species replacement of aquatic plant community in east lake. Acta Phytoecol Sin 21:319–327

Ye C, Yu HC, Kong HN, Song XF, Zou GY, Xu QJ, Liu J (2009) Community allocation of four submerged macrophytes on two kinds of sediments in Lake Taihu, China. Ecol Eng 35:1656–1663

Yu FH, Wang N, He WM, Dong M (2010) Effects of clonal integration on species composition and biomass of sand dune communities. Biochemistry 40:15707–15715

Zhang M et al (2011) Light-dependent phosphate uptake of a submersed macrophyte *Myriophyllum spicatum* L. Aquat Bot 94:151–157

Zhang Q et al (2012) Does light heterogeneity affect structure and biomass of submerged macrophyte communities? Bot Stud 53:377–385

Zhang Q, Xu YS, Huang L, Xue W, Sun GQ, Zhang MX, Yu FH (2014) Does mechanical disturbance affect the performance and species composition of submerged macrophyte communities? Sci Rep 4:254

Zhu G, Zhang M, Cao T, Ni L, Zhong A, Fu H (2012) Effects of sediment type on stem mechanical properties of the submerged macrophyte *Hydrilla verticillata* (L.F.) royle. Fresen Environ Bull 21:468–474